Guilherme Braga

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Lista 1 → as derivadas de Backpropagation "na mão"

(c)

$$\nabla_\theta J(\theta) = \left(\frac{dJ}{dw_1}, \dots, \frac{dJ}{db_3}\right)$$

$\left\{ \begin{array}{l} 6 \text{ pesos, } w_1, \dots, w_6 \\ 3 \text{ bias } b_1, b_2, b_3 \end{array} \right.$

→ a regra da cadeia é

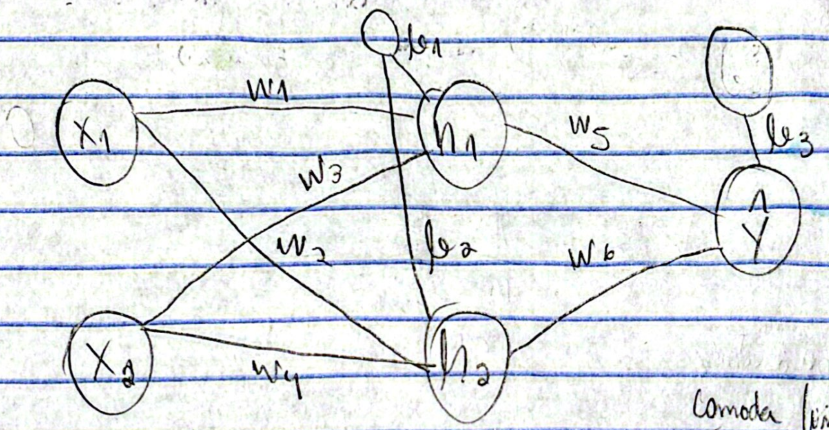
$$\boxed{\frac{d}{dx} f(g(x)) = \frac{df}{dg} * \frac{dg}{dx}}$$

sigmoide $\sigma(x) = \frac{1}{1+e^{-x}}$

↓

derivada $\sigma'(x) = \sigma(x) * (1-\sigma(x))$

"Fazendo Backpropagation"



Camada final "2"

camada escondida "1"

$$\Rightarrow \frac{dJ_i}{dw_5} = \frac{dJ_i}{d\hat{y}_i} * h_{1i} \xrightarrow{\text{logo}} \frac{dJ_1}{dh_1} = \frac{dJ_i}{d\hat{y}_i} * w$$ (I)

$$\Rightarrow \frac{dJ_i}{dw_6} = \frac{dJ_i}{d\hat{y}_i} * h_{2i} \xrightarrow{\text{logo}} \frac{dJ_1}{dh_2} = \frac{dJ_1}{d\hat{y}_1} *$$ (II)

Camada do meio $\left\{ \begin{array}{l} a_{1i} = x_{1i}w_1 + x_{2i}w_3 + b_1 \\ a_{2i} = x_{1i}w_2 + x_{2i}w_4 + b_2 \end{array} \right.$

$\Downarrow$

$h_{ki} = \sigma(a_{ki})$

$\Downarrow$

$$\Rightarrow \frac{dJ_i}{db_3} = \frac{dJ_i}{d\hat{y}_i} = -2(y_i - \hat{y}_i) = 2(\hat{y}_i - y_i$$

em h1 $\frac{dJ_i}{da_{1i}} = \frac{dJ_i}{dh_{1i}} \sigma'(a_{1i})$

(I) ↳ $\frac{dJ_i}{d\hat{y}_i} * w_5 * h_{1i}(1-h_{1i})$ (IV) em h_2: $\frac{dJ_1}{da_{2i}} = \frac{dJ_1}{dh_{2i}} \sigma'(a_{2i})$

$= \frac{dJ_i}{d\hat{y}_i} * w_4 * h_{2i} * (1-h_{2i})$ V

de IV e V retiro

$\frac{dJ_i}{dw_1} = \text{IV} * x_{1i}$ | $\frac{dJ_i}{dw_2} = \text{V} * x_{1i}$

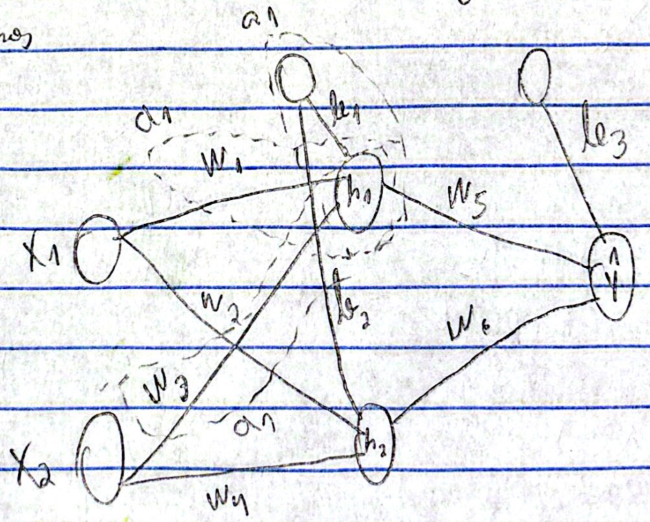
$\frac{dJ_i}{dw_3} = \text{IV} * x_{2i}$ | $\frac{dJ_i}{dw_4} = \text{V} * x_{2i}$

$\frac{dJ_1}{db_1} = \text{IV}$ | $\frac{dJ_i}{db_2} = \text{V}$

$$\frac{dJ}{d\theta_j} = \nabla J_\theta = \frac{1}{m} \sum_{i=1}^{m} \frac{dJ}{d\theta_j}$$

↳ parâmetros

pois



Logo se é W_1 é só a conta de a_1 por X_1 como em cima!